

Common Computing Devices -- Guided Notes ANSWER KEY

Unit tp-2.1 | CompTIA Network+ N10-009 Obj. FC0-U71 2.1 | N10-008 FC0-U71 2.1

For instructor use / distribute after students complete the notes.

Question 1

A _____ is a full-powered computer with separate components (tower, monitor, keyboard) designed for a fixed location. A _____ integrates all components into a portable unit with a built-in screen and battery. A _____ is a portable touchscreen device, typically without a physical keyboard.

Answer: Desktop; laptop; tablet.

Question 2

A _____ is a purpose-built computer that provides resources (files, print, web, database) to many client devices over a network. Servers typically have multiple _____ (processors), large amounts of _____, and enterprise-grade reliability features such as ECC memory and RAID storage.

Answer: Server; CPUs (processors); RAM.

Question 3

A _____ (IoT device) is any internet-connected device that is not a traditional computer, such as a smart thermostat, IP camera, or wearable. These devices typically have _____ (limited/unlimited) processing power and often run _____ operating systems with minimal user interaction.

Answer: Smart device (IoT); limited; embedded.

Question 4

A thin client is a low-powered device that relies on a _____ server to do most processing. The thin client handles _____ (input/output on the screen, keyboard/mouse) but the actual computation and storage happen on the server. This model reduces _____ and simplifies updates.

Answer: Remote; input/output (display and interaction); hardware cost.

Question 5

A workstation is a high-performance desktop designed for resource-intensive tasks such as _____, _____, or _____ (name at least two). Workstations typically have ECC RAM, professional-grade _____, and may support multiple monitors.

Answer: Video editing, 3D rendering, CAD/engineering design, scientific simulation; GPU (graphics card).

Question 6

Match each device to its primary use case:

Laptop: _____

Tablet: _____

Desktop: _____

Server: _____

Answer: Laptop: mobile productivity; Tablet: content consumption, touchscreen interaction; Desktop: stationary, upgradeable workstation; Server: providing services to multiple clients.

Question 7

A _____ is a handheld computer that combines phone and computing functions, running a mobile OS such as Android or iOS. It includes sensors such as _____, _____, and _____ that traditional desktops lack, enabling location services, motion detection, and photography.

Answer: Smartphone; GPS, accelerometer, camera (any three sensors).

Question 8 -- Real World Application

REAL WORLD: A small dental office is choosing between deploying thin clients connected to a central server vs. individual laptops for each workstation. List two advantages of thin clients for this scenario and two advantages of laptops, then explain which solution you would recommend and why.

Answer: Thin client advantages: lower hardware cost per station; centralized management (software updates and backups happen on the server, not each device); easier to secure because patient data never sits on individual machines. Laptop advantages: work continues if the server goes down; staff can work from outside the office; each device is independently functional. Recommendation: thin clients are better for a dental office because patient records (HIPAA-protected data) should be stored centrally and never on portable devices that could be lost or stolen. Centralized backups also reduce the risk of data loss.